Growth & Remodeling of Soft Tissue

A hands-on introduction to the four modelling frameworks

kinematic growth · constrained mixture · homogenized · equilibrated

teacher version — key equations filled in

Reproducible figures generated by the `gr` package – `github.com/YaleCBL-Teaching/Growth-Remodeling`

Roadmap

Part 1 — foundations

1. The biology: why tissues grow & remodel
2. Two mechanical stimuli: pressure & flow
3. Finite-strain continuum mechanics
4. Kinematic growth

Part 2 — mixture theories & outlook

5. Constrained mixture (full)
6. Homogenized constrained mixture
7. Equilibrated constrained mixture
8. Stability: the one big question
9. Future perspectives
10. Open-source implementations

One big question: *when does a loaded tissue adapt to a stable new state, and when does it grow without bound?*

Interactive throughout: 🛠 "Your turn" analytical & coding exercises every few slides.

1. The biology

why living tissues grow and remodel

Living materials are never “finished”

A steel beam is made once and wears out. A living artery is **continuously torn down and rebuilt**: cells deposit and degrade structural proteins while sensing the loads they carry.

- **Growth** — change in *mass*.
- **Remodeling** — change in *microstructure* (fiber orientation, natural length, composition).

Will a hypertensive artery thicken and stabilise, or an aneurysm dilate to rupture?

The cast of characters in an arterial wall

Constituent	Mechanical role	Turnover in adults
Elastin	soft; bears load at low stretch	**none** — half-life of decades
Collagen	stiff, crimped; guards against rupture	**fast** — weeks to months
Smooth muscle	passive load + active contraction; the sensor cell	fast; switches phenotype

- **Elastin cannot be renewed** — its loss triggers *aneurysm*.
- **Collagen & SMC turn over constantly** — so the wall can *adapt*, or *run away*.

Source: Humphrey, Eberth, Dye, Gleason, J Biomech (2009)

Deposition pre-stretch

Cells cross-link each new fiber *while the wall is loaded*, so it is **born pre-stretched** to the same target state. Each cohort keeps its *own* natural configuration — a **constrained mixture**: deforming together now, each remembering a different stress-free state.

no deposition pre-stress

New material enters slack ($\sigma_{\text{pre}} = 0$); as old fibers are removed, the replacement carries no load — the homeostatic state **cannot be maintained**.

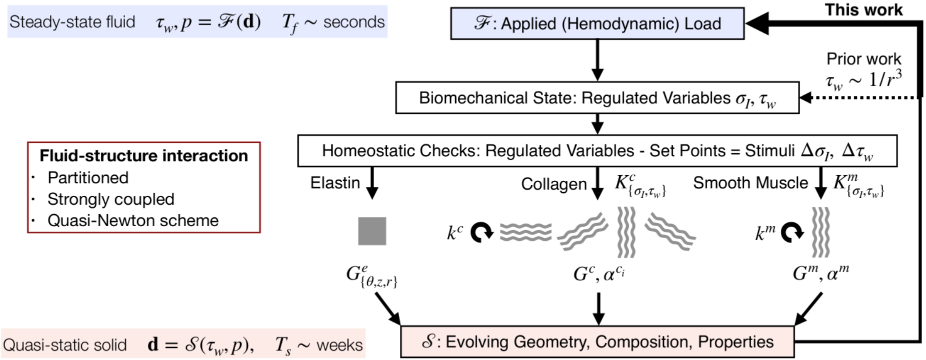
with deposition pre-stress

New material enters at the homeostatic prestress $\sigma_{\text{pre}} = \sigma_h$ (stretch G^k); each replacement immediately bears its share — the state is **maintained through turnover**.

This one convention is what lets a continuously-renewed tissue hold a steady mechanical state.

Cells chase a mechanical set-point

Cells regulate a **target (homeostatic) stress**: above it they build more, below it removal wins — a negative feedback that thickens a hypertensive wall until stress returns to normal.

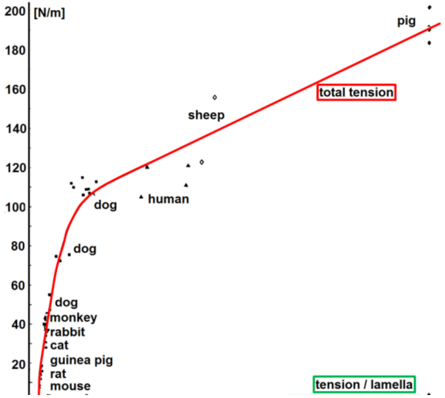


Regulated variables are compared to set-points; the mismatch drives constituent-specific G&R.

Adapted from Humphrey, J Elasticity (2021)

Fundamental observation: a stress set-point

Wolinsky & Glagov (1967): Cauchy stress in the aortic media is nearly **constant across mammals** spanning orders of magnitude in body mass — direct evidence of a mechanical set-point. (Davis 1867 & Wolff 1892: tissue adds material along sustained tension; Grossman 1975: pressure overload thickens the wall just enough to normalise stress.)



Wall tension vs. body mass across species.

Two canonical scenarios we will simulate

Hypertension — sustained pressure step. Expect *adaptation*: the wall thickens, stress returns to set-point.

Aneurysm — irreversible elastin loss. Geometry can feed *positively* back on stress: stabilise larger, *or* dilate without bound.

When does a tissue adapt, and when does it run away?

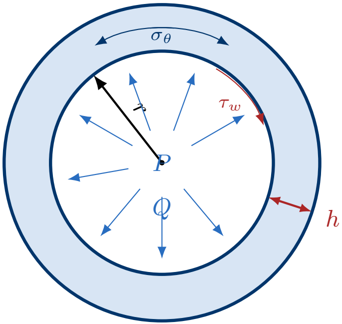
The four theories answer this differently — and the code lets you explore each.

2. Two mechanical stimuli

pressure sets thickness, flow sets radius

An idealized blood vessel

A vessel is a **pressurised, perfused tube**. Two loads act on the wall: **blood pressure** P pushes it outward, and **blood flow** Q drags its inner surface. Cells sense both.



Source: Humphrey, Cardiovascular Solid Mechanics (2002)

- P lumen pressure $\rightarrow$ **circumferential stress** σ_θ
- Q volumetric flow $\rightarrow$ **wall shear stress** τ_w
- r inner radius, h wall thickness

Cells regulate *both* σ_θ and τ_w toward set-points — by changing h and r.

Circumferential stress — the Laplace law

Balancing lumen pressure against wall tension for a thin-walled tube gives the **circumferential (hoop) stress**:

$$\boxed{\sigma_\theta = \frac{P\,r}{h}} \tag{1}$$

Thicker wall $\Rightarrow$ lower stress; larger radius or pressure $\Rightarrow$ higher stress. This is the load the intramural cells (SMC, fibroblasts) feel.

Source: Laplace law; Humphrey, Cardiovascular Solid Mechanics (2002)

Wall shear stress — the flow the endothelium feels

For steady laminar (Poiseuille) flow Q of a fluid with viscosity μ through a tube of radius r, the shear stress the flowing blood exerts on the wall is

$$\boxed{\tau_w = \frac{4\mu\, Q}{\pi\, r^3}} \tag{2}$$

The endothelium senses τ_w; note the steep r^{-3} dependence — a small radius change moves shear stress a lot.

Source: Poiseuille flow; Humphrey, Cardiovascular Solid Mechanics (2002)

Two homeostatic set-points, two adaptations

Each stimulus has a preferred value the cells defend. Rearranging the two laws shows *what geometry restores each set-point*:

Pressure $\to$ thickness

Hold $\sigma_\theta = \sigma_h$:

$$h = \frac{P\,r}{\sigma_h} \;\propto\; P.$$

Higher pressure $\Rightarrow$ **thicker wall**.

Flow $\to$ radius

Hold $\tau_w = \tau_h$:

$$r = \left(\frac{4\mu\,Q}{\pi\,\tau_h}\right)^{1/3} \propto\; Q^{1/3}.$$

Higher flow $\Rightarrow$ **larger lumen**.

These two rules are the mechanobiology in miniature — and we can explore them analytically before writing any code.

Source: Humphrey, Cardiovascular Solid Mechanics (2002)

Your turn: Pressure adaptation

Question. A patient becomes hypertensive: pressure rises to $P' = (1+\gamma)P$, but the radius r is held fixed. By how much must the wall thicken to bring the circumferential stress back to its set-point σ_h? *pen & paper*

Resources. The Laplace law (1); homeostatic stress σ_h.

Hints. Write σ_θ before and after; set the after-value equal to σ_h and solve for the new thickness h'.

Take a minute — the answer is on the next slide.

Answer: Pressure adaptation

Initially homeostatic: $\sigma_h = Pr/h$. After the step, demanding $\sigma_\theta = P'r/h' = \sigma_h$ with $P' = (1+\gamma)P$ gives

$$\boxed{h' = (1+\gamma)\,h.} \tag{3}$$

Reading. The wall thickens *in proportion* to the pressure rise — exactly Grossman's observation, and the outcome a healthy hypertensive adaptation should reproduce. Radius unchanged, stress restored.

Source: Grossman, Jones, McLaurin, J Clin Invest (1975)

Your turn: Flow adaptation

Question. Chronically increased flow raises $Q' = (1+\varepsilon)Q$. By how much must the lumen radius change to restore the wall-shear-stress set-point τ_h? *pen & paper*

Resources. The wall-shear-stress law (2); set-point τ_h.

Hints. Only r adapts. Set $\tau_w(Q', r') = \tau_h$ and use $\tau_h = 4\mu Q/(\pi r^3)$ to eliminate constants.

The r^{-3} dependence makes this one quick.

Answer: Flow adaptation

From $\tau_h = 4\mu Q'/(\pi r'^3) = 4\mu Q/(\pi r^3)$, the radius ratio is $(r'/r)^3 = Q'/Q = 1+\varepsilon$, so

$$\boxed{r' = (1+\varepsilon)^{1/3}\, r.} \tag{4}$$

Reading. The lumen enlarges as the *cube root* of the flow increase — the classic flow-induced outward remodeling. A $\sim$30% flow rise widens the vessel by only $\sim$9%.

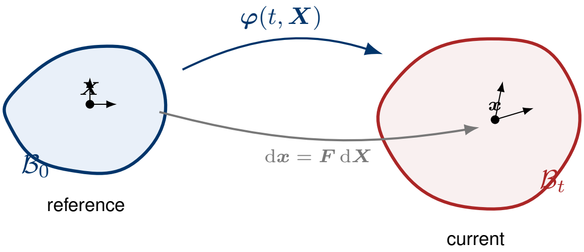
Source: Kamiya & Togawa, Am J Physiol (1980); Humphrey, Cardiovascular Solid Mechanics (2002)

3. Finite-strain fundamentals

configuration, deformation, strain, stress, hyperelasticity

Reference and current configuration

A body is described in two configurations: the **reference** $\mathcal{B}_0$ (material points $\boldsymbol{X}$) and the **current** $\mathcal{B}_t$ (spatial points $\boldsymbol{x}$). The **motion** $\boldsymbol{x} = \boldsymbol{\varphi}(t, \boldsymbol{X})$ carries one to the other.



Source: continuum-mechanics background adapted from YaleCBL-Teaching/BENG_5570

The deformation gradient

How a line element transforms is the key kinematic quantity — the **deformation gradient** $\boldsymbol{F} = \partial \boldsymbol{x}/\partial \boldsymbol{X}$ maps reference line elements to current ones:

$$\boxed{\mathrm{d}\boldsymbol{x} = \boldsymbol{F}\,\mathrm{d}\boldsymbol{X}, \qquad \boldsymbol{F} = \frac{\partial \boldsymbol{\varphi}}{\partial \boldsymbol{X}}} \tag{5}$$

Its determinant $J = \det \boldsymbol{F}$ is the volume ratio; **elastic incompressibility** means $J_e = 1$ (only growth changes volume). Reducing to one dominant direction, $\boldsymbol{F}$ collapses to a single scalar **stretch**

$$\lambda = \frac{\text{current length}}{\text{reference length}}, \qquad \lambda = 1 \text{ undeformed.}$$

Strain — a rotation-free measure of stretching

$\boldsymbol{F}$ still contains rigid rotation. Removing it gives the **right Cauchy–Green** tensor and the **Green–Lagrange strain**:

$$\boxed{\boldsymbol{C} = \boldsymbol{F}^\top \boldsymbol{F}, \qquad \boldsymbol{E} = \tfrac{1}{2}(\boldsymbol{C} - \boldsymbol{I})} \tag{6}$$

$\boldsymbol{E} = \boldsymbol{0}$ at no deformation. In 1D, $C = \lambda^2$ and $E = \frac{1}{2}(\lambda^2 - 1)$ — and the fiber invariant $I_4 = \lambda_e^2$ used below is exactly C read along a fiber.

Stress — force per current area

We report the **Cauchy (true) stress** σ: force per unit *current* area — the physically meaningful measure for a growing body, because the load-bearing area itself changes as the tissue remodels. It is what enters the Laplace law (1).

- Reference-area measures (1st/2nd Piola–Kirchhoff) exist too, but for a remodeling tissue “per current area” is the natural bookkeeping.
- Balance of linear momentum in the current configuration reduces, for our quasi-static thin-walled vessel, to exactly the Laplace relation.

Hyperelasticity: stress from a stored-energy function

A **hyperelastic** material stores energy $W(\lambda_e)$ and its stress is the derivative — no dissipation, path-independent:

$$\boxed{\sigma \;=\; \lambda_e \, \frac{\mathrm{d}W}{\mathrm{d}\lambda_e}} \tag{7}$$

Elastin — neo-Hookean $W_e = c_e(I_1 - 3)$ (soft, barely stiffening):

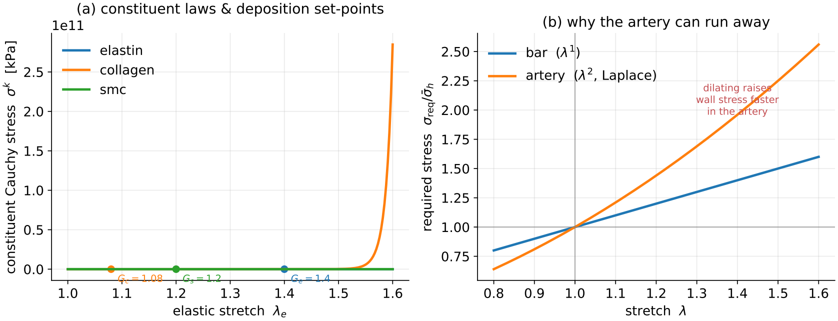
$$\sigma_e(\lambda_e) = c_e\left(\lambda_e^2 - \tfrac{1}{\lambda_e}\right).$$

Collagen & SMC — Fung fiber (crimped, then stiffens steeply):

$$\boxed{\sigma(\lambda_e) = c_1\, \lambda_e^2(\lambda_e^2 - 1)\, e^{\,c_2(\lambda_e^2-1)^2}, \qquad I_4 = \lambda_e^2} \tag{8}$$

Source: constituent laws as in Latorre, Humphrey, Comput Methods Appl Mech Eng (2020)

Constituent laws & why the artery is special



Left: soft elastin vs. steeply stiffening collagen/muscle, with G^k and σ_h^k marked. Right: the required stress grows *linearly* for the bar but *quadratically* for the artery (Laplace).

That extra power of λ is the whole story of arterial instability (§8).

Reproduced with the gr package (this repo)

The deposition stretch G^k

Cells deposit new fibers **already under tension** at a fixed deposition stretch $G^k > 1$. At homeostasis the elastic stretch equals G^k, which *defines* the homeostatic stress:

$$\boxed{\sigma_h^k := \sigma^k(G^k)} \tag{9}$$

This one convention pins down every set-point in the course — so all four theories share one parameter set and compare like-for-like.

Source: deposition-stretch concept: Humphrey, Rajagopal, Math Models Methods Appl Sci (2002)

Multiplicative decomposition — growth vs. elasticity

Split the deformation into a stress-free **inelastic** part (growth / remodeling) and an **elastic** part — the only part that carries stress:

$$\boxed{\lambda = \lambda_e \, \lambda_{\text{inel}}} \tag{10}$$

- **Kinematic growth** (§4): *one* split, an inelastic growth stretch.
- **Constrained mixtures** (§5): each cohort its own natural configuration, superposed.

The set-point (9) and the split (10) recur in every model that follows.

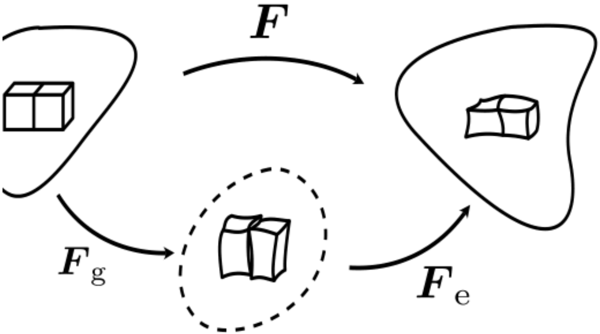
4. Kinematic growth

Rodriguez, Hoger & McCulloch (1994)

The idea, and the kinematics

Postulate a stress-free **growth** deformation; only the elastic part carries load. Growth is one scalar θ (radial thickening):

$$\boldsymbol{F} = \boldsymbol{F}_e\,\boldsymbol{F}_g, \qquad \frac{M}{M_0} = \theta, \qquad \text{only } \boldsymbol{F}_e \text{ stores energy.}$$



One growth variable for the whole mixture — no individual turnover. That is the strength and the ceiling

The growth law

Growth nulls the stress deviation from the set-point (first-order, production $\propto$ current mass):

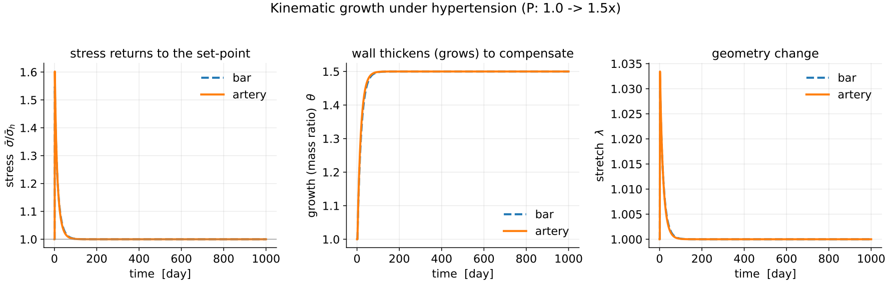
$$\boxed{\frac{\mathrm{d}\theta}{\mathrm{d}t} = k_g\,\theta\left(\frac{\bar{\sigma}}{\bar{\sigma}_h} - 1\right)} \tag{11}$$

The set-point $\bar{\sigma}_h$ is **prescribed**: wherever growth stops ($\dot{\theta} = 0$), stress is homeostatic *by construction*. At each instant λ follows from equilibrium $\bar{\sigma}(\lambda) = \sigma_{\text{req}}(\lambda, \theta)$.

Source: Rodriguez, Hoger, McCulloch, J Biomech (1994)

Stability is imposed, not predicted

Growth is stable iff adding mass *lowers* the stress it must carry, $\mathrm{d}\bar{\sigma}/\mathrm{d}\theta|_{\theta^*} < 0$ — always true for the **bar**, but a strong enough insult flips the sign for the **artery** (Laplace λ^2) and growth runs away.



Kinematic growth either converges to the *prescribed* set-point or diverges — it cannot *predict* stability from turnover.

Reproduced with the `gr` package (this repo)

Your turn: Kinematic growth — code

Question. Drive the same tissue and watch it either reach the *prescribed* set-point or run away. *ex02*

Resources. `gr` package, `exercises/ex02_kinematic_growth.py`. Tunables: `K_G=0.05`, `PRESSURE_FACTOR=1.5`, `GEOMETRY="artery"/"bar"`.

Hints. Watch the stress plateau land on $\bar{\sigma}_h$. Then crank `PRESSURE_FACTOR` on the `artery` until it diverges — the `bar` never does.

Run. `uv run python exercises/ex02_kinematic_growth.py`

Answer: Kinematic growth

The stress plateau *always* lands on the set-point — stability is **imposed**, not predicted. The bar is unconditionally stable; the artery (Laplace λ^2) tips into runaway once `PRESSURE_FACTOR` is large enough.

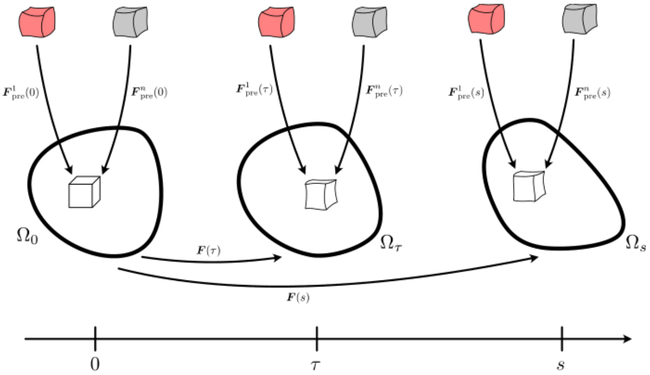
Learning. Kinematic growth reaches the a-priori prescribed homeostasis, *or* it is unstable — nothing in between.

5. Constrained mixture (full)

Humphrey & Rajagopal (2002) — the heart of the lecture

The idea: track every cohort

Constituents are **constrained to deform together**, but each carries its *own* natural configuration and is continuously **produced and removed**. The tissue therefore carries a whole *history* of cohorts — faithful, and expensive.



Material deposited at $0, \tau, s$ is laid down at the deposition stretch G^k into successive configurations.

Fig.: A. Gebauer (TUM lecture); Humphrey, Rajagopal, Math Models Methods Appl Sci (2002)

Mass balance — the heredity integral

The mass of a turnover constituent k now is the surviving remnant of the initial material plus everything produced since, each aged by a survival function:

$$\boxed{M^k(t) = M^k(0)\, q^k(t,0) + \int_0^t m^k(\tau)\, q^k(t,\tau)\,\mathrm{d}\tau} \tag{12}$$

Survival (Poisson removal at rate k_d^k):

$$q^k(t,\tau) = e^{-k_d^k (t-\tau)}.$$

Mean lifetime $1/k_d^k$; elastin does not turn over ($k_d^e = 0$).

Source: Humphrey, Rajagopal, Math Models Methods Appl Sci (2002)

Production $\propto$ current mass, driven by the tissue-stress stimulus that vanishes at homeostasis:

$$m^k = k_d^k M^k\, \Upsilon, \quad \Upsilon = 1 + K_\sigma^k \left(\frac{\bar{\sigma}}{\bar{\sigma}_h} - 1\right).$$

Each cohort remembers its birth configuration

A cohort of k deposited at time τ is laid down at the deposition stretch G^k; as the tissue goes on deforming, that cohort stretches *with* the mixture, so its elastic stretch evaluated now is

$$\boxed{\lambda_e^k(t;\tau) = G^k \, \frac{\lambda(t)}{\lambda(\tau)}} \tag{13}$$

At birth ($t = \tau$) it sits exactly at G^k (hence σ_h^k). This is the **constrained-mixture constraint**: every constituent shares the current deformation, but each cohort measures it from a different starting point.

Source: Humphrey, Rajagopal, Math Models Methods Appl Sci (2002)

Mixture stress = history integral

The mixture Cauchy stress is the mass-weighted sum over *all* surviving cohorts (rule of mixtures):

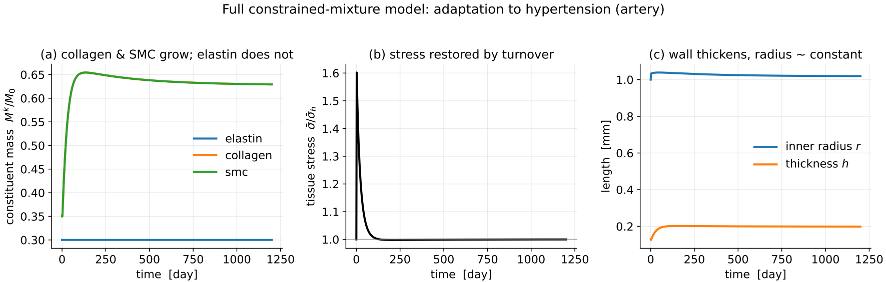
$$\boxed{\bar{\sigma}(t) = \frac{1}{M_{\text{tot}}} \sum_k \left[M^k(0)\, q^k(t,0)\, \sigma^k\big(G^k \lambda(t)\big) + \int_0^t m^k(\tau)\, q^k(t,\tau)\, \sigma^k\Big(G^k \tfrac{\lambda(t)}{\lambda(\tau)}\Big)\, \mathrm{d}\tau \right]} \tag{14}$$

Each time step discretises the mass balance and this sum over stored cohorts, then solves $\bar{\sigma}(\lambda) = \sigma_{\text{req}}(\lambda)$ for the current stretch. Elastin contributes a single, non-renewing cohort (optionally degraded by an insult).

Source: Humphrey, Rajagopal, Math Models Methods Appl Sci (2002)

What it buys you: stability is *predicted*

Nothing prescribes the outcome — adaptation or dilation **emerges** from the turnover dynamics and the insult.



Under hypertension collagen and muscle grow, elastin is diluted, and tissue stress returns to homeostatic.

The price: history integrals make the model $O(N^2)$ in time steps — the motivation for the homogenized model (§6).

Reproduced with the gr package (this repo)

Your turn: Constrained mixture — code

Question. Change how fast the tissue turns over and how hard it responds; compare hypertension against an elastin-loss (aneurysm) insult. *ex03*

Resources. `exercises/ex03_constrained_mixture.py`. Tunables: `TURNOVER_DAYS=20`, `GAIN=1.0`; `INSULT = ` hypertension (`pressure_factor=1.5`) vs. aneurysm (`elastin_surviving=0.3`).

Hints. Try turnover 10 / 20 / 70 d and gain 0.3 / 1.0 / 3.0. Does the same tissue adapt to one insult but not the other?

Run. `uv run python exercises/ex03_constrained_mixture.py`

Answer: Constrained mixture

Mild hypertension **adapts**: collagen and SMC grow, tissue stress returns to homeostatic. A severe elastin insult can **run away**. Faster turnover / higher gain reach the new state sooner.

Learning. Stability is *predicted*, and it **depends on the insult** — not fixed in advance.

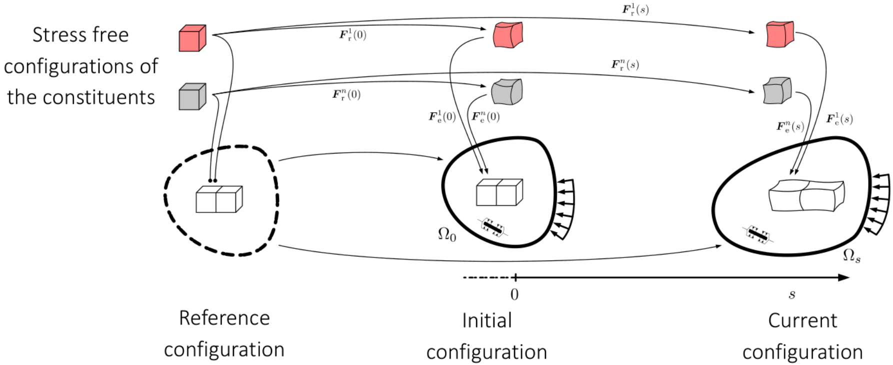
6. Homogenized CMM

Cyron, Aydin & Humphrey (2016)

Throw away the history, keep its average

Temporal homogenization replaces the entire deposition history by a *single evolving mean natural configuration*. The heredity integral becomes **two ODEs per constituent**; cost drops from $O(N^2)$ to $O(N)$:

$$\lambda_e^k = \frac{\lambda}{\lambda_n^k}.$$



One mean natural configuration per constituent, evolving from reference through initial to current state

Growth (mass) and remodeling (natural configuration)

Growth — mass production $\propto$ current mass, driven by the tissue-stress deviation:

$$\boxed{\frac{\mathrm{d}M^k}{\mathrm{d}t} = M^k \left(K_\sigma^k k_d^k\right) \left(\frac{\bar{\sigma}}{\bar{\sigma}_h} - 1\right)} \tag{15}$$

Since production scales with current mass, $\dot{M}^k = 0 \Leftrightarrow \bar{\sigma} = \bar{\sigma}_h$ for *any* mass — mass settles wherever it must to restore tissue stress.

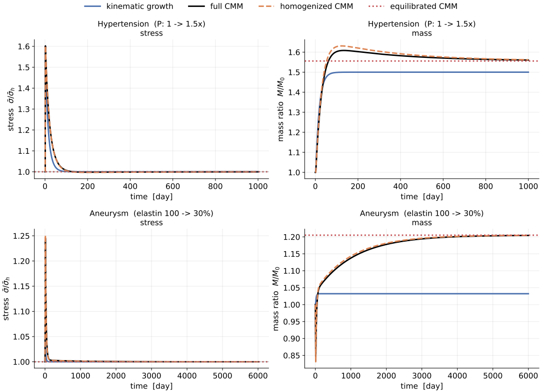
Remodeling — new mass enters at λ/G^k, old mass leaves at the current λ_n^k; mixing at the turnover rate,

$$\boxed{\frac{\mathrm{d}\lambda_n^k}{\mathrm{d}t} = \frac{m^k}{M^k} \left(\frac{\lambda}{G^k} - \lambda_n^k\right)} \tag{16}$$

Holding λ fixed, this relaxes constituent stress exponentially to σ_h^k (time constant $\sim 1/k_d$) — Cyron's **mechanical analog**: a Maxwell element in parallel with a motor holding the homeostatic prestress.

Source: Cyron, Aydin, Humphrey, Biomech Model Mechanobiol (2016)

It tracks the full model closely — much cheaper



Homogenized (dashed orange) sits almost on the full CMM (black); kinematic growth (blue) under-responds to elastin loss — it grows constituents in lockstep and cannot remodel them individually.

Learning: the homogenized model follows the full model's *time trajectory* closely.

Reproduced with the `gr` package (this repo)

Your turn: Homogenized vs. full — code

Question. Overlay the homogenized model on the full constrained mixture for the same insult, and measure both the agreement and the speed-up. *ex04*

Resources. `exercises/ex04_homogenized_vs_full.py`. Tunables: `INSULT` (hypertension / aneurysm), `T_END=2000`.

Hints. Read off the gap between the two curves. Try the aneurysm insult or a longer `T_END` — does the agreement hold?

Run. `uv run python exercises/ex04_homogenized_vs_full.py`

Answer: Homogenized vs. full

The homogenized curve overlays the full CMM closely across the transient, at a fraction of the runtime ($O(N)$ vs. $O(N^2)$).

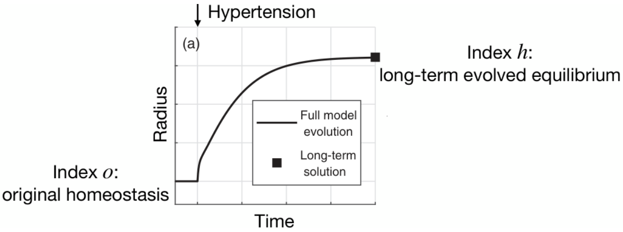
Learning. The homogenized model reproduces a **similar time trajectory** to the full model — cheaply.

7. Equilibrated CMM

Latorre & Humphrey (2018)

Skip the transient — solve for the end state

Set every rate to zero and solve the **algebraic** system — one root-find instead of thousands of time steps. At mechanobiological equilibrium: (A) growth balance $\Upsilon = 1 \Rightarrow \bar{\sigma}^* = \bar{\sigma}_h$; (B) remodeling done $\lambda_e^k = G^k \Rightarrow \sigma^k = \sigma_h^k$; (C) mechanical equilibrium at the evolved geometry.



The equilibrated model jumps straight to the long-term evolved state (index h) that the transient theories approach in time.

Fig.: M. Pfaller (BMES 2023); Latorre, Humphrey, ZAMM (2018)

Collapse to one scalar equation

Let the turnover constituents grow in fixed proportion (exact when they share gain and turnover). Conditions (A)–(C) reduce to a *single* equation for the evolved stretch λ^*:

$$\boxed{\phi_{e0}\, s + s\, \frac{\bar{\sigma}_h - \sigma_e(G_e \lambda^*)}{\sigma_{\text{turn}} - \sigma_h^e} - \gamma\, (\lambda^*)^n = 0} \tag{17}$$

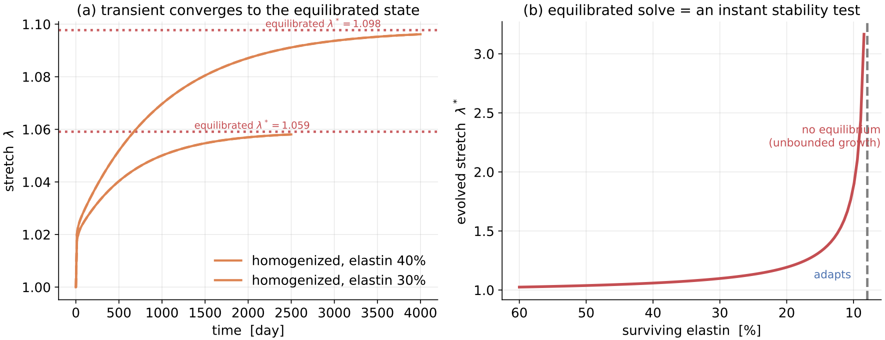
with n the geometry exponent (1 bar, 2 artery), γ the sustained load factor, s the surviving elastin fraction. Once λ^* is known, the evolved mass ratio is $M/M_0 = \gamma(\lambda^*)^n$.

Because (6.1) is just the fixed point of the homogenized ODEs, the equilibrated solution **coincides with the long-time limit of the transient theories** — to several digits.

Source: Latorre, Humphrey, ZAMM (2018)

The punchline: existence = stability

Equation (17) may have **no physical root** — then no adapted state exists and the transient theories dilate without bound. The solver is therefore the cleanest **stability test**: root $\Rightarrow$ adapts, no root $\Rightarrow$ unstable.



Equilibrated solution matches the transient end-state where a root exists; near the boundary the transient slows critically.

Learning: the equilibrated model matches the transient theories *iff* an equilibrium exists.

Reproduced with the gr package (this repo)

Your turn: Equilibrated — code

Question. Check that the algebraic equilibrium matches the transient end-state — then find the elastin loss at which the equilibrium *disappears*. *ex05*

Resources. `exercises/ex05_equilibrated.py`. Tunable: `ELASTIN_SURVIVING=0.3` (try 0.4 / 0.3 / 0.2).

Hints. Part A: the equilibrated λ^* should sit on the transient plateau. Part B: lower s until the root vanishes — that is s_{crit}; run just below it and watch the transient diverge.

Run. `uv run python exercises/ex05_equilibrated.py`

Answer: Equilibrated

Where a root exists, the equilibrated solve lands exactly on the transient end-state. Below s_{crit} there is *no* root — and the transient dilates without bound.

Learning. The equilibrated model **matches** the transient theories *if* an equilibrium exists — and flags, by failing to find one, exactly when it does not.

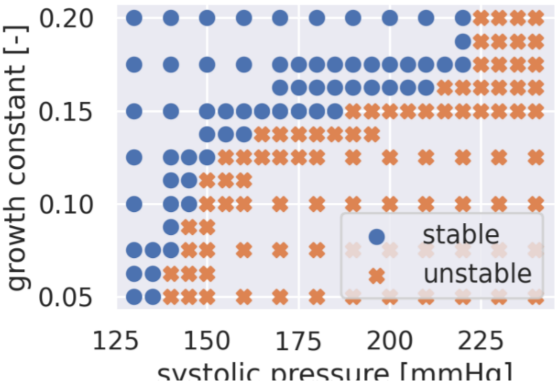
8. Stability

adaptation vs. aneurysm — the one big question

Where instability comes from: the Laplace exponent

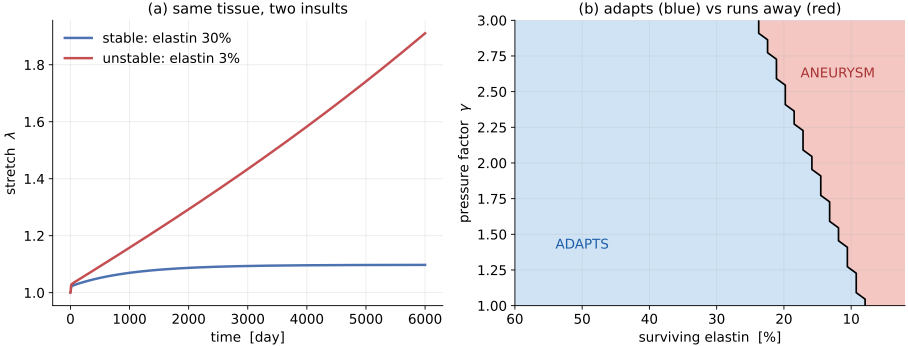
Growth adds mass, *lowering* required stress (stabilising); dilation *raises* it, **quadratically** for the artery. Lose elastin and dilation wins the race — the feedback turns positive: **runaway**.

$$\boxed{\left.\frac{\mathrm{d}\bar{\sigma}}{\mathrm{d}\theta}\right|_{\theta^*} < 0 \;\Leftrightarrow\; \text{tissue adapts}} \tag{18}$$



Same tissue, two insults

Read stability two ways: **existence** — the equilibrated equation (17) has a root iff the tissue can adapt (no time integration); or **transient** — integrate and watch it plateau or climb.



Retaining 30% elastin the artery adapts (blue); retaining 3% it dilates without bound (red). Map computed from the equilibrated existence test.

Reproduced with the gr package (this repo)

How the four theories answer — the takeaways

Theory	Predicts stability?	What it says
Kinematic growth	No — *imposed*	Reaches the *prescribed* homeostatic set-point, or runs away. Stability is built in, not learned from turnover.
Constrained mixture (full)	Yes	Adapts or dilates **depending on the insult** — the faithful reference.
Homogenized CMM	Yes	Same verdict as the full model, following its **time trajectory** cheaply.
Equilibrated CMM	Yes — *instantly*	**Matches** the transient theories *if* a root exists; *no* root $\Rightarrow$ unbounded growth.

Stability *depends on the insult*, and — with turnover-based theories — it can be *predicted*, not just assumed.

Your turn: Stability map — code

Question. Reproduce the stable/unstable map, then change *one* tissue property and watch the "aneurysm" region move. *ex06*

Resources. `exercises/ex06_stability_and_aneurysm.py`. Tunables: `GAIN=1.0`→3.0, `COLLAGEN_C1=250`→600, `COLLAGEN_G=1.08`→1.15.

Hints. Each change shifts the boundary between adaptation and runaway. Which property shrinks the unstable region most?

Run. `uv run python exercises/ex06_stability_and_aneurysm.py`

Answer: Stability map

Higher gain, stiffer collagen, or larger deposition stretch all enlarge the *adapts* region — the tissue tolerates a stronger insult before running away.

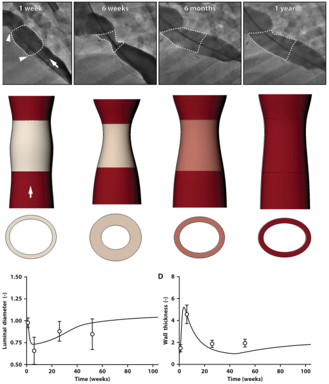
Learning. The same existence test that defines the equilibrium boundary *is* the stability criterion.

9. Future perspectives

clinical translation, systems biology, fluid–solid–growth

Clinical application: reversible graft stenosis

A constrained-mixture G&R model predicted — and in-vivo imaging confirmed — that early narrowing of tissue-engineered vascular grafts can **spontaneously reverse** as neotissue matures. Much observed “failure” is transient.

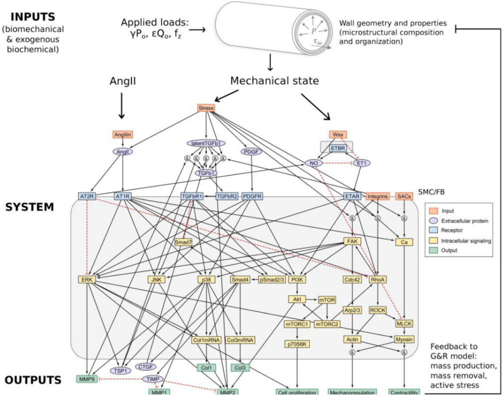


Predicted and observed reversal of TEVG stenosis over time.

Drews, Pepper, Best, . . . , Humphrey, Breuer, Sci Transl Med (2020)

Systems-biology coupling: signaling to remodeling

The stress stimuli here are *phenomenological*. A next step couples the tissue-level model to a **logic-based cell-signaling network** — mechanics drives signaling (AngII, TGF-β, . . .), whose outputs feed back on mass production and active stress: “from transcript to tissue”.

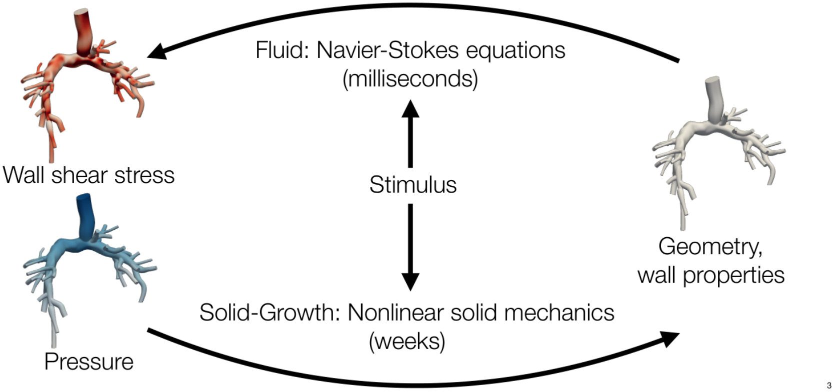


Intracellular signaling network linking mechanical stimuli to constituent turnover.

Irons, Latorre, Humphrey, Ann Biomed Eng (2021)

Fluid–solid–growth: closing the hemodynamic loop

In a real vessel, G&R changes geometry $\to$ flow $\to$ wall shear stress $\to$ the remodeling stimulus. **FSG** couples the fast fluid (Navier–Stokes, ms) to the slow solid-growth (weeks); **FSGe** makes it fast by using the *equilibrated* solver (§7) for the solid.

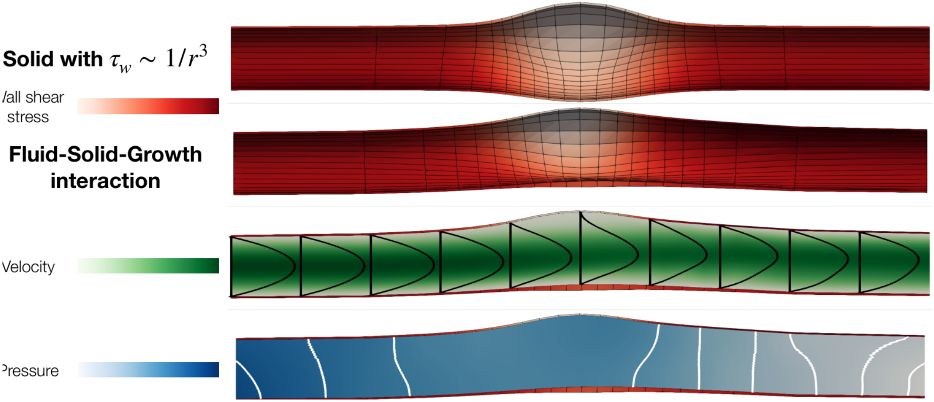


The fluid–solid–growth feedback loop across disparate timescales.

Fig.: M. Pfaller (TUM/Stanford talk)

FSGe: long-term evolved aneurysmal equilibrium

Coupling the equilibrated solid to the hemodynamics yields the evolved geometry *and* the flow fields at the adapted state — far cheaper than a fully transient simulation. The equilibrated theory is the enabling ingredient for tractable 3D FSG.



Evolved aneurysm: wall shear stress, velocity, and pressure fields at the long-term equilibrium.

Pfaller, Latorre, Schwarz, Gerosa, Szafron, Humphrey, Marsden, Comput Methods Appl Mech Eng (2024)

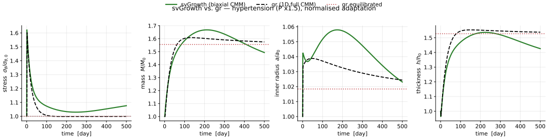
10. Open-source implementations

from teaching code to research solvers

This seminar's code: a reproducible workflow

Everything you have seen is generated by the `gr` Python package in **one repository**:

- all four theories on *one* parameter set and *one* mechanical setting — apples-to-apples;
- deliberately reduced-order (1D / thin-membrane, single stress component) so it runs in seconds;
- `docs/` lecture notes, `exercises/` student stubs, `solutions/` that regenerate every figure.



Reproduced with the `gr` package + svGrowth (this repo)

`make figures` in `slides/` regenerates fig01–06 from the code, so the deck never drifts from the model.

Source: github.com/YaleCBL-Teaching/Growth-Remodeling

Research-grade open-source solvers

Project	What it does	Stack / group
svGrowth	constrained-mixture G&R of vessels (biaxial thin-wall); the cross-validation reference used above	Python; Stanford CBCL
svFSGe	fast, strongly-coupled 3D fluid–solid–growth driver (the FSGe method)	Python + svFSI/svMultiPhysics; Stanford CBCL
FEMBE_Plugin	mechanobiologically equilibrated constrained-mixture material as a FEBio plugin	C++ / FEBio; Yale (Humphrey lab)
FEFSG_Plugin	fluid–solid–growth coupling as a FEBio plugin (thoracic-aorta hypertension & aneurysm cases)	C++ / FEBio; Yale (Humphrey lab)

`github.com/StanfordCBCL/svGrowth` `github.com/StanfordCBCL/svFSGe`

`github.com/yale-humphrey-lab/FEMBE_Plugin` `github.com/yale-humphrey-lab/FEFSG_Plugin`

Source: svGrowth builds on Humphrey, Rajagopal, Math Models Methods Appl Sci (2002); FEMBE on Latorre, Humphrey, Comput Methods Appl Mech Eng (2020)

Summary — what to take away

1. Living tissue continuously **turns over**; cells deposit new material *pre-stretched*, chasing a homeostatic mechanical set-point.
2. **Kinematic growth** imposes that set-point — simple and robust, but stability is assumed, not predicted.
3. **Constrained mixtures** predict stability from turnover, and it **depends on the insult**: mild $\rightarrow$ adapt, severe $\rightarrow$ runaway.
4. The **homogenized** model reproduces the full model's time trajectory at a fraction of the cost.
5. The **equilibrated** model matches the transient theories *when an equilibrium exists*, and flags instability by failing to find one.
6. These tools now drive **clinical prediction** (TEVGs), **systems-biology coupling**, and **fluid–solid–growth**.

When does a loaded tissue adapt, and when does it run away? — now you can compute the answer.

Key references I

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Key references II

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Key references III

Code

- Seminar code (this deck's figures): `github.com/YaleCBL-Teaching/Growth-Remodeling`
- `github.com/StanfordCBCL/svGrowth`, `svFSGe`
- `github.com/yale-humphrey-lab/FEMBE_Plugin`, `FEFSG_Plugin`

Figure sources. Schematic and hand-annotated figures adapted from lecture material by A. Gebauer (TUM) and prior talks by M. Pfaller; result figures reproduced with the `gr` package; literature figures cited per-slide.